

Akash Singh

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EDUCATION

Birla Institute of Technology, Mesra

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Ranchi, India

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript (ES6+), C, SQL, HTML5, CSS3

Frontend Development: React.js, Next.js 16, React 19, Tailwind CSS v4, Zustand, Framer Motion, Vite

Backend & Real-Time: Node.js, Express.js, Socket.IO (WebSockets), REST APIs, SSR, Better Auth, Clerk, JWT

Databases & Storage: MongoDB, Mongoose ODM, Geospatial Indexing, Aggregation Pipelines, LocalStorage

Tools & Cloud Platforms: Git, GitHub, Postman, Vercel, Render, Stripe API, npm, Bun, VS Code, Cursor

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, OS, Computer Networks

FEATURED PROJECTS

SnapCart | *Next.js 16, React 19, TypeScript, Socket.IO, MongoDB, Stripe, Tailwind CSS*

[GitHub](#) | [Live Demo](#)

- Architected a full-stack, production-grade grocery delivery platform with 3 isolated role-based portals for **Customers, Delivery Partners, and Admins**.
- Implemented real-time bidirectional communication using **Socket.IO** for live courier GPS tracking, instant status broadcasts, and customer-rider in-app messaging.
- Integrated **Stripe Payment Gateway** with webhooks for automated order lifecycle management and implemented secure OTP delivery verification for fraud prevention.

Chai Charcha | *Next.js 16, React 19, TypeScript, MongoDB, Better Auth, Tailwind CSS*

[GitHub](#) | [Live Demo](#)

- Engineered a community forum application featuring topic-based guilds, nested multi-level discussion threads, and real-time karma reputation voting.
- Built granular **Role-Based Access Control (Admin/Moderator/Member)** with automated audit logging, content moderation queues, and user report resolution.
- Optimized database queries with compound indexing and aggregation pipelines, reducing post feed latency by 40%.

Eventra | *React.js, Node.js, Express.js, MongoDB, QR Code Engine, Tailwind CSS*

[GitHub](#) | [Live Demo](#)

- Developed a campus event management and ticketing platform supporting interactive event browsing, schedule calendars, and seat availability tracking.
- Built an **Admin Check-in Dashboard** featuring camera-based live QR code verification for offline-capable, tamper-proof ticket scanning.
- Hardened backend endpoints with unique database indexing and transactional capacity checks to prevent double-booking under concurrency.

Prescripto | *React.js, Node.js, Express.js, MongoDB, RESTful APIs, Tailwind CSS*

[GitHub](#) | [Live Demo](#)

- Developed a full-stack medical appointment scheduling platform with doctor specialization filtering, dynamic date slot availability, and patient booking records.
- Designed clean REST APIs with Express.js and MongoDB to manage appointment state transitions and physician schedules.

ACHIEVEMENTS & CODING PROFILE

Problem Solving: Solved **200+** **Data Structures & Algorithms** problems in **Java** on LeetCode focusing on Graphs, Dynamic Programming, Trees, and Arrays.

Open Source & Shipping: Published and actively maintain **30+** **repositories** on GitHub with production deployments on Vercel and Render.

Hackathons: Participated in the **UIDAI Hackathon** and engineered **AtharvaAI** (AI-powered cultural knowledge comparison engine).